Translating survey records into tax-return terms

John Ricco

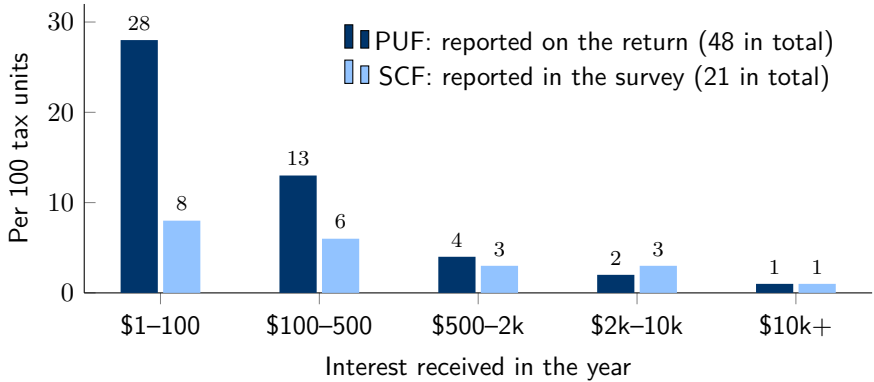
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The problem

- The model is estimated on the PUF and four surveys together: the SCF, CEX, CPS and SIPP. Variables that appear in several sources, such as wages, interest and Social Security, are estimated from all of them.
- The surveys do not measure these variables the way a tax return does.
  - Definitions differ. CPS wages are before pre-tax deductions such as 401(k) contributions.
  - Amounts are top-coded or misreported.
  - Small amounts often go unreported.
- Without a correction, the model would learn a mixture of the survey's concept and the tax return's. Variables found only in surveys, such as wealth, would be estimated conditional on the wrong amounts.

Example: interest income



The gap is in small amounts, which many households do not report in the survey. Above $2,000 the two sources agree.

Married couples with wages of $75,000–200,000. 2015 PUF and 2016 SCF (tax year 2015), weighted.

Two steps

1. **A mapping function for each survey variable.** We look at how the survey's variable differs from the tax return's and choose a mapping by hand. It is applied once, before the model is estimated.

2. **A fill-in for units that report zero.** Some units that report zero in the survey would show an amount on a tax return. The fill-in assigns amounts to some of them. It is applied while the model is estimated.

Both steps compare each survey with the tax data for the same year: the PUF's filers, grown to that year, plus the non-filers from the CPS.

Step 1: mapping functions

Kind of difference	Mapping	Example
None	Keep the reported dollars	SCF interest and dividends
A known difference in definition	Apply the tax rule	SCF capital losses held to the $3,000 a return allows
The order of units is right but the dollar scale is not	Percentile among holders	SCF and CPS wages; CPS dividends
Reliable for small amounts, not at the top	Keep the dollars below a line, percentile above it	SIPP dividends, with a line at $50,000

Step 1 example: wages by percentile

- CPS wages include pre-tax deductions, and the CPS has more high earners than tax returns show. In 2019 it has 29.0 million units with wages above $100,000, against 21.9 million in the tax data.
- The mapping keeps the CPS's ranking of earners and takes the dollar amounts from the tax data.
- A CPS unit at the 90th percentile among CPS earners is given the wages of the unit at the 90th percentile among earners in the tax data.
- The assumption is that the survey ranks earners correctly even where its dollar amounts are off.

Units with no wages are handled by step 2.

Step 2: two versions of each filled-in variable

SCF household	interest, as reported	interest, as on a return
household 1	$250	$250
household 2	$0	?
household 3	$0	?
household 4	$1,800	$1,800

- The survey answer is kept and never changed.
- The tax-return version equals the answer for units that report an amount. For units that report zero it is drawn.
- PUF records have only the tax-return version. The model learns the tax-return version from all sources.

Step 2: assumptions

Within a type of unit (married or not, number of dependents, wage band):

1. The tax data gives the true share of units with an amount in each size range.
2. A unit that reports an amount reports it correctly. Only zeros can be wrong.
3. Whether a holder reports depends only on its type and the size of its amount. Two couples of the same type with \$50 of interest are equally likely to leave it out, whatever else is on their records.

Under these assumptions, the reporting rate for size range k is the survey's share over the tax data's share, $r_k = B_k/A_k$. In the interest example, for \$1–100 it is $8/28 = 0.29$.

Step 2: the chance a zero-reporter holds an amount

For a zero-reporter with record x, Bayes' rule gives the chance that it holds an amount in range k:

$$\Pr(\text{holds } k \mid \text{reports } 0, x) = \frac{\Pr(\text{holds } k \mid x)\,(1 - r_k)}{\Pr(\text{reports } 0 \mid x)} = \frac{C_k(x)\left(A_k/B_k - 1\right)}{C_0(x)}$$

- $C_k(x)$ is the model's probability that a unit with record x reports an amount in range k. $C_0(x)$ is its probability of reporting zero.
- By assumption 3, $\Pr(\text{holds } k \mid x) = C_k(x)/r_k$, which gives the second equality.
- $A_k/B_k - 1$ is the number of hidden holders per reporter in the unit's type. It sets how many amounts are filled in.
- $C_k(x)$ sets which units receive them: those whose other answers look like reporters'.

A_k: the tax data's share in range k, by type. B_k: the average of $C_k(x)$ over the survey's units of that type. The total chance is capped at 100 percent.

Step 2 example: interest for married couples

Per 100 couples	Tax data	SCF	Filled in
$1–100	28	8	20
$100–500	13	6	7
$500–2k	4	3	2
$2k and over	3	3	0
Any	48	21	28

A typical couple of this type, for $1–100:

- $C_1 = 0.08$ and $C_0 = 0.79$
- hidden holders per reporter: $28/8 - 1 = 2.5$
- chance: $0.08 \times 2.5/0.79 = 0.25$

A couple with large savings and stock holdings might have $C_1 = 0.20$ and $C_0 = 0.55$. Its chance is $0.20 \times 2.5/0.55 = 0.91$.

Married couples with wages of $75,000–200,000; 2015 PUF and 2016 SCF, weighted. The second couple's probabilities are made up for illustration.

Step 2 within pretraining

1. **Passes 1 and 2.** No amounts are filled in.
2. **At the start of each later pass**, for every filled-in variable and survey:
   - A_k by type, from the tax data for the survey's year;
   - $C_k(x)$ for every survey unit, from the model's current parameters, with the tax-return versions hidden;
   - B_k, the average of $C_k(x)$ within each type;
   - each zero-reporter's chance of holding an amount in each range.
3. **In each batch**, new amounts are drawn for the zero-reporters from these chances. The parameters are then updated on the tax-return version.

The chances use only survey answers and the tax data, never earlier fill-ins. As the model improves, it gets better at telling which zero-reporters look like holders.

Checking the result

Share of SIPP units with none, 2019, during the 28 October run:

	As reported	After the fill-in	Tax data
Wages	30.5%	21.2%	22.5%
Dividends	86.0%	79.6%	80.4%

- A report compares each variable as reported, after step 1, after step 2 and in the tax data, for every survey and year.
- The fill-in is applied to wages, interest, dividends, capital gains, pensions, business income and Schedule E, among others, in each survey that measures them.

The tax-data figures are taken at the SIPP's mix of household types.